

Real Mode	Protected Mode	Q. Real mode vs Protected Mode
Segment Offset	Selector Descriptor	Q. Global vs Local descriptors
1MB		Total descriptors (both local & global 16,384)

Protected Mode

- X No segment register
- ✓ Operates above the first 1MB allocated real memory
- ✓ Used in Windows operating system.
- ✓ "Selector" selects a descriptor

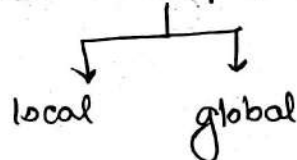
[Segment Register]



Selector



Select descriptor from descriptor table



Contents of descriptor

- ↳ Segment location
- ↳ Length
- ↳ Access Rights

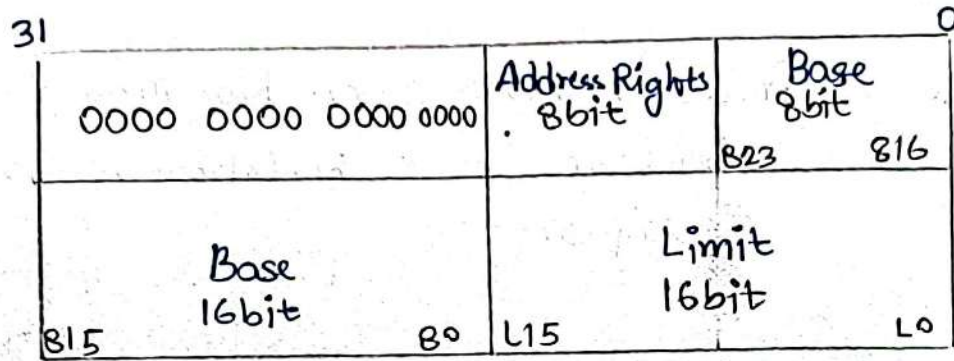
[Selector is located in the segment register]

→ Selects one of 8192 descriptors from 1/2 descriptor tables

Local descriptors are unique to application

Global descriptor ~ System Descriptor

Local descriptor ~ Application Descriptor



$$\text{Base} = 16 \text{ bit} + 8 \text{ bit} = 24 \text{ bit}$$

$$\text{Limit} = 16 \text{ bit}$$

~~Access~~

$$\text{Access right} = 8 \text{ bit}$$

table 2 [Page 64]

Figure 2-6

Core 2 → each descriptor is 8 bytes in length.

Maximum in descriptor table = 64K

Base address denotes starting location.

→ 24 bits

Bits → Granularity bit

If $A=0$,

If $A=1$,

→ Example 2.1

$$\text{Base} = \text{Start} = 10000004$$

$$A = 0$$

$$\text{End} = \text{Base} + \text{Limit} = 10000004 + 001FFH$$

$$= 10001FFH$$

Example 2.2

$$G = 1$$

Base = same as before

$$\text{End} = \text{Base} + \text{Limit} (\text{append} + \text{FFF})$$

$$= 1000000H + 001FFFFF$$

$$= 101FFFFFH$$

$$\text{Bits} = L \ \& \ AV$$

$$\begin{cases} L=1 \rightarrow 64 \text{ bit} \\ L=0 \rightarrow 32 \text{ bit} \rightarrow \text{compatibility mode} \end{cases} \rightarrow 64 \text{ bit descriptor}$$

AV $\sim$ is segment available

AV = 0 $\rightarrow$ unavailable

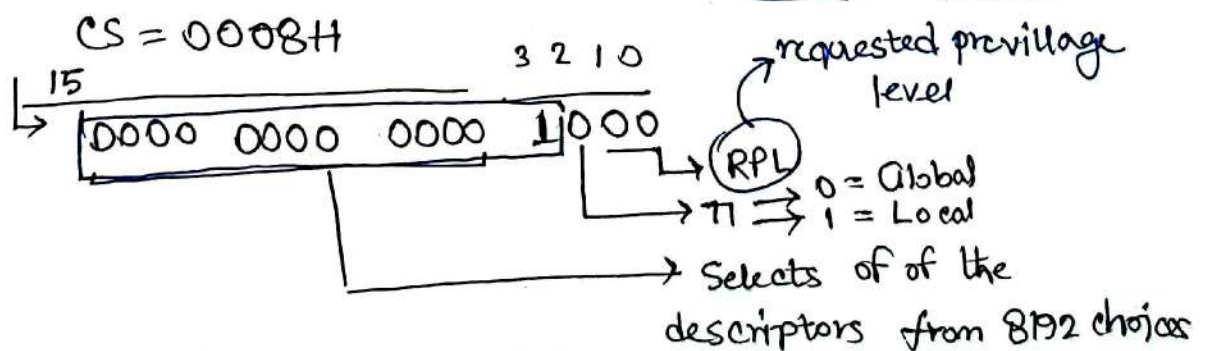
AV = 1 $\rightarrow$ segment available

$$\text{Bits} = D$$

D = 0 $\rightarrow$ 16 bit instruction

D = 1 $\rightarrow$ 32 bit instruction

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Shayari

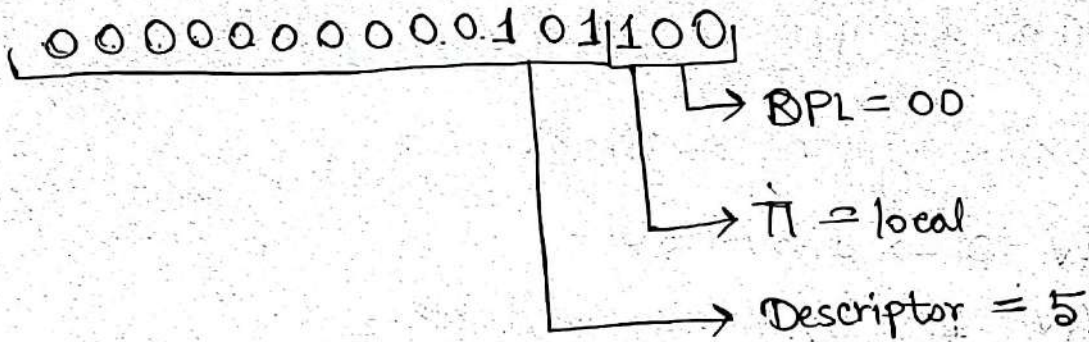
DP L

DP L

RPL = 00
DPL = 01

☒ RPL approved

Class Work

$$CS = 002CH$$


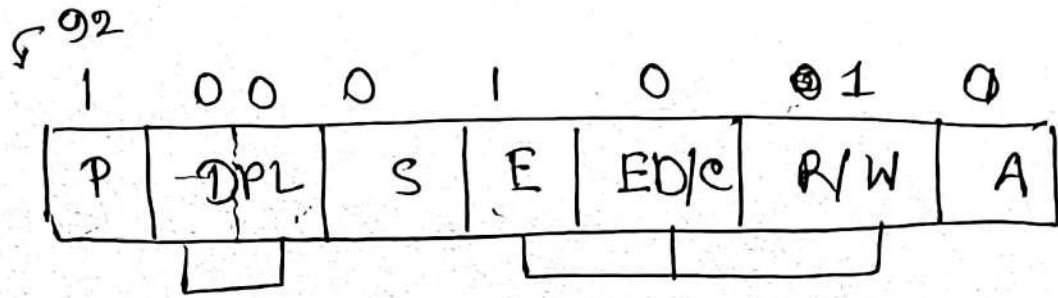
~ Descriptor ~

00 00 92 10 00 00 00 FF

0000	92	10
0000	00FF	

~~82006~~
80206

Shayan



13 selector chooses → descriptor

→ if the requested privilege level matches or higher in priority than ...

Microcontroller — 2 set questions
Microprocessor — 2 set questions